As a pathologist, I studied disease through a microscope.

As an epidemiologist, I studied patterns across communities.

Then it became personal.

After I lost my job, my family lost our health insurance. The next month, my husband was bitten by a tick and became ill with Lyme disease.

I understood disease. But as a wife, I faced another question: How do we find our way through this?

For immigrants like my husband, there can be another barrier: language. It can be hard to feel comfortable explaining symptoms to a doctor who does not speak your own language.

That is why I built PathwayAI.

I'm Yuwei Zhang. Let me show you.

For the TOPx HHS Cost of Illness challenge, PathwayAI looks beyond the medical bill. It helps patients prepare for care and helps county and community teams identify practical support priorities.

The pathway starts before a bite, with prevention. After a bite, it explains removal and when to seek medical advice.

But knowing you need care is only the beginning.

What if you don't have insurance?

Here, people can find community health centers serving uninsured patients, ask about income-based fees, and explore coverage or hospital financial assistance. Care is not automatically free, but there is a place to start. No story is required to use these resources.

What if you struggle to explain what happened?

On both the tick-bite and care pages, people can enter notes in their preferred language. With permission, AI translates and organizes them into an English visit summary, shown alongside the original.

The person can review, correct, and download it. They can also prepare a request for a qualified interpreter.

This is preparation for a conversation, not a replacement for medical interpretation. Our aim is to make communication easier, so language is less likely to stand between someone and care.

Now imagine a person preparing for an appointment. This example is fictional.

They describe possible exposure, a tick encounter, symptoms that began two weeks later, and a blood test they report was positive.

The summary keeps that sequence visible. It separates symptom onset from diagnosis timing, preserves reported test results, and leaves unknown information unknown.

A clinician still needs the original test report and a clinical assessment. The patient brings a clearer account they can reuse across appointments.

Now look at the burden around that care.

This person paid two hundred twenty-five dollars for care and travel. They missed three workdays. A caregiver contributed eight hours.

Money paid. Time lost. Help needed.

PathwayAI keeps those measures separate. Someone facing a bill can record the balance and prepare assistance questions. Someone needing transport can prepare questions about eligibility, cost, booking, and the return trip.

For community teams, these experiences raise a practical question: Where could support make a difference?

The app brings together public county evidence, published cost research, and transparent planning scenarios. Dutchess and Orange provide the starting comparison.

These illustrations show possible scale, not measured county spending or the full long-term burden.

A planning brief helps partners discuss existing services, staffing, costs, and possible funding. Identifying a gap does not create resources, so the response must be feasible.

Then follow up: Did people reach care? Did assistance come through? Who still needs help?

Shared patient-report collection is currently off. A future consented pilot would need secure follow-up to understand how burdens change over time.

AI helps people communicate. Transparent calculations explain costs. Patients and community partners decide what matters.

Early treatment can help prevent serious complications. PathwayAI aims to help people reach care and support sooner. Reduced delays, disability, and costs remain outcomes we need to evaluate.

That is PathwayAI: Full Economic Burden.

Making the Hidden Patient Journey Visible.

To patients, caregivers, and experts: thank you. Your experiences and guidance helped make this possible.